Technical Proposal: Cookie Squad (TrustGuard, Track 2)

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Our workflow (pink = data work, where most of our gain came from)

1. Clean	2. Build features	3. Score	4. Blend	5. Adjust	6. Flag
fix missing, invalid values	relative amounts, thresholds, combos	three logistic regressions	weights 56 / 24 / 20%	shift groups that differ in test	top ~2.5% score 0.5+

1. Problem Understanding

The task is to score each transaction by how likely it is to be fraud. Only **353 of the 20,000** training transactions are fraud (1.77%), so a model that always answers "legitimate" is 98% accurate and catches nothing. We therefore optimised **PR-AUC**, which measures how well frauds are ranked above legitimate transactions; a random ranking scores roughly the fraud rate, about 0.02. F1 and recall depend on the yes/no cut-off, which should match how many cases a review team can check, since a missed fraud loses money and a false alarm blocks a real customer.

2. Data Processing & Feature Engineering

Inputs are cleaned first (missing columns added as blanks, invalid numbers blanked, values clipped to the training range, category text standardised) so unusual rows in the hidden test set cannot crash the notebook. Four findings from exploring the data shaped the features:

- **Amounts only mean something in context.** The median `luxury` transaction is about 847 while the median `food_delivery` one is about 30. We therefore compare each amount (on a log scale) with the median for its category, channel and country, and with the account's own average over the last 24 hours (`spend_last_24h` / `transactions_last_24h`).
- **Risk only rises past a threshold.** Fraud became more likely above about 2 transactions in the last hour or 9 in the last 24 hours, so we count only the excess above those levels. We also flag bursts (16 or more transactions in 24 hours) and accounts younger than 120 days.
- **Some signals only matter together.** `new_device` is common and harmless on an old account but risky on a new one or late at night, so we added **new device × night** and **new device × account age**.
- **The test set is distributed differently from training.** Bursts are 3.1 times as common in the test set, high-value purchases on old accounts about 4 times, and missing values rise from 0.52% to 1.81%. About 3,000 test rows also look machine-generated: their values are spread evenly within fixed ranges and none occur at 5 to 7am or 9 to 10pm.

Finally, a **nearest-neighbour feature** measures how close each transaction is to the nearest known fraud compared with the nearest known legitimate transaction from training, so the model can recognise cases that resemble past frauds.

3. Model Selection & Justification

We combine three **logistic regressions**, each fed different features: one on all the features above (56% of the final score), one on every pair of 11 core features multiplied together so it can learn combinations (24%), and one on yes/no steps such as "amount over 500" or "account younger than 365 days", because generated data tends to use hard cut-offs (20%). Each is L2 regularised (large weights are penalised) so it cannot memorise 353 examples; the pairwise model, with 147 weights, is penalised most. Gradient boosting, CatBoost, neural networks and TabPFN all scored lower, likely because 353 frauds are too few for them. It assumes that each feature adds or removes a fixed amount of risk on the log-odds scale (which is why we built the combinations and thresholds by hand), that each transaction can be judged on its own (there is no account history), and that the fraud patterns in training still apply in the test set.

Because the test set differs, we then **shift the scores of specific groups**. A shift of −2.0 divides a group's odds of fraud by about 7, +1.0 multiplies them by about 2.7, and the order within the group never changes. Bursts on accounts up to a year old get −2.0 and on older accounts −0.3, since many test bursts behave like ordinary heavy users; bursts from a country other than Malaysia, Indonesia or Singapore then get +0.5 back (70 test rows). New devices on accounts over a year old get −0.75. Purchases of 2,000 to 6,000 in `luxury`, `cash_transfer`, `electronics` or `travel` on accounts over a year old get +1.0, and 1,000 to 2,000 get +0.5. The 975 rows of the largest machine-generated block are all moved close to a 3.5% score, keeping 30% of their original differences, because individual scores there appear to be noise.

4. Evaluation Strategy

Model choices were tested with **stratified 5-fold cross-validation, repeated with 3 different random splits**. Each fold keeps the 1.77% fraud rate, and label-based values were recomputed inside each fold so validation rows never saw their own answers. The blended model scores a PR-AUC of **0.251** this way. The group shifts target test-set differences that training data cannot show, so each was kept only if it raised the score on the public leaderboard. Each stage of our public score:

What we added	Public PR-AUC
One logistic regression on our earlier, smaller feature set	0.172
New-device combinations, amount compared with channel and country, nearest-neighbour feature	0.186
Group shifts for the test set (bursts, new devices, high-value purchases, machine-generated block)	0.223
Second model (pairwise combinations) blended in	0.233
Third model (yes/no thresholds) blended in	0.234
Burst shift relaxed for countries other than Malaysia, Indonesia and Singapore	**0.242**

Some of the data related stages add about 0.06 of the total 0.07 gain; the two extra models add 0.011. For F1 and recall, we add the same constant to every score so that about the top 2.5% of transactions (307 of the 12,000 in the public test) score above 0.5; the ranking and PR-AUC are unchanged.

5. Limitations & Risks

The group shifts were tuned with **public-leaderboard feedback**, so they may not transfer if the hidden test set has a different mix; we kept them few and simple, and each only affects rows matching its rule. The 3.5% level is a tuned setting, not a measured fraud rate, and the view that most test bursts are legitimate is our interpretation, not confirmed by labels. Outputs are ranking scores, not calibrated probabilities. Next steps: account history, cost-based cut-offs and drift monitoring.